Feature Engineering — v4

Chest X-Ray Multi-Condition Detection — DataML Challenge

Update covering feature set H (adds a new vessel/rib-density family, via a Frangi vesselness filter, to the fourteen families in feature set G)

1. Why hand-crafted features

The challenge rules do not allow neural networks or pretrained models, so the model cannot learn its own features from raw pixels the way a CNN would. Instead, every feature is written by hand using classical image processing (NumPy, OpenCV, scikit-image, SciPy, PyWavelets). Each feature is a formula that turns one X-ray image into a small set of numbers describing one specific property of the image — brightness, texture, or shape.

Fifteen feature families have now been designed in total (up from fourteen in the previous version), each capturing one specific property of the image. The notebook groups them into feature sets of increasing size, A through H. Feature set H is now the active setting, and includes all fifteen families: intensity, spatial, texture, gradient, morphology, symmetry, wavelet, frequency, blob, ROI, GLRLM, Gabor, Hu moments, clinical/anatomical, and vessel/rib density — 231 raw numeric features per X-ray image (up from 225 under set G).

2. Feature families used (feature set H)

Every family is computed on the same preprocessed image, so the comparison between families is fair. Family O is new in this version.

Family	#Feat.	What it measures	Why included	Disease sign targeted
A. Intensity	31	Overall brightness of the image: global stats, percentiles, a 16-bin brightness histogram.	Diseased tissue often looks denser (whiter) or hazier than normal air-filled lung, shifting overall brightness stats.	Infiltration, Effusion
B. Spatial grid	48	3x3 grid + top/middle/bottom zones; brightness stats per cell/zone.	Tells the model where in the chest something looks unusual, not just whether the whole image looks unusual.	Effusion (lower), Pneumothorax (upper)
C. Texture (LBP+GLCM)	30	Local pattern of the image: smooth, patchy, or repeating.	Some conditions create patchy/irregular tissue patterns instead of a smooth lung field.	Infiltration, Nodule
D. Gradient/edge	12	How sharp and in what direction intensity changes are, plus edge density.	Some conditions produce a sharp boundary line a normal lung field does not have.	Pneumothorax, Nodule
E. Morphology (shape)	8	Shape stats of bright/dark regions: count, size, roundness, solidity.	A round, solid, well-defined shape is a basic sign of a nodule.	Nodule
F. Symmetry	5	Compares left half of the chest to the right half (brightness, histogram, edge density).	The two lungs are normally close to mirror images; many conditions affect only one side.	Pneumothorax, Effusion
G. Wavelet	28	Multi-resolution (DWT, db4, level 2) detail decomposition: LL approximation + LH/HL/HH detail bands per level.	Captures fine, localized texture at multiple zoom levels that single-scale texture stats miss.	Nodule, Infiltration
H. Frequency (FFT)	4	Ratio of image energy across low/mid/high spatial-frequency radial bands + spectral entropy.	Diffuse haze concentrates energy at low frequencies; small sharp structures push energy higher.	Infiltration, Nodule

Family	#Feat.	What it measures	Why included	Disease sign targeted
I. Blob/lesion	5	Detects small round blob-like structures via Laplacian-of-Gaussian (blob_log).	Targets round, spot-like lesions directly instead of inferring shape indirectly through morphology.	Nodule
J. ROI-aware	7	Repeats brightness + left/right stats restricted to the segmented lung mask only.	Removes rib cage, spine, and background contamination from whole-image averages.	Effusion, Pneumothorax
K. GLRLM	10	Gray-Level Run-Length Matrix: length of runs of similar intensity, horizontal & vertical.	Short high-intensity runs are a proxy for nodular texture vs. the long smooth runs typical of normal lung parenchyma.	Nodule, Infiltration
L. Gabor	24	Multi-frequency (3), multi-orientation (4) Gabor filter response: mean + std per combination.	Captures oriented, fine-grained texture that plain LBP/GLCM can underrepresent.	Infiltration, Atelectasis
M. Hu moments	7	7 rotation/scale/translation-invariant moment descriptors of the lung mask shape (log-scaled).	Captures gross distortion of the lung silhouette — e.g. a collapsed lung (Atelectasis) or a flattened diaphragm line (Effusion).	Atelectasis, Effusion
N. Clinical / Anatomical	6	Costophrenic-angle sharpness (per side) + gross left-right lung shape/volume ratios, computed directly on the lung mask.	Targets two specific radiological signs directly — costophrenic-angle blunting and diaphragm flattening/lung-volume distortion.	Effusion (blunted angle), Atelectasis (shape/volume distortion)
O. Vessel / Rib Density (new)	6	Frangi vesselness-filter density of tubular structures (vessels, bronchial markings, ribs), overall and left vs. right.	A different signal family from texture roughness (LBP/GLCM/Gabor) — targets vascular/rib markings directly, including whether they look crowded, obscured, or asymmetric.	Infiltration (vessel obscuration), Atelectasis (vessel crowding)
Total (feature set H)	**231**	Combined raw feature count extracted per image, before cleaning.		

3. Full list of features, by family

The detailed breakdown behind the counts in Section 2 — actual column names produced by extract_all_features() for feature set H. Numbered suffixes index histogram bins, grid cells, GLCM angle/distance combinations, wavelet levels, or Gabor frequency/orientation combinations, rather than being separate hand-picked features. Families A–N are unchanged from feature set G; Family O is new.

A. Intensity (31 features)

`Global distribution stats (9): int_mean, int_median, int_std, int_var, int_min, int_max, int_skew, int_kurt, int_entropy`

`Percentiles (6): int_p5, int_p25, int_p50, int_p75, int_p95, int_iqr`

`Brightness histogram (16): int_hist_0 … int_hist_15`

B. Spatial grid (48 features)

`3×3 grid cells (36): grid_{row}_{col}_mean, grid_{row}_{col}_std, grid_{row}_{col}_p75, grid_{row}_{col}_entropy`

Top/middle/bottom zones (12): zone_top_mean, zone_top_std, zone_top_p75, zone_top_entropy, zone_mid_(…), zone_bottom_(…)

C. Texture — LBP + GLCM (30 features)

Local Binary Pattern histogram (18): lbp_s1_0 … lbp_s1_17

GLCM stats (12): glcm_s1_contrast_mean/std, glcm_s1_dissimilarity_mean/std, glcm_s1_homogeneity_mean/std, glcm_s1_energy_mean/std, glcm_s1_correlation_mean/std, glcm_s1_ASM_mean/std

Note: single-scale only (s1) under feature set H — see Section 9 limitation.

D. Gradient / edge (12 features)

Gradient magnitude & edge density (4): grad_mag_mean, grad_mag_std, grad_mag_p95, edge_density

Gradient orientation histogram (8): grad_orient_hist_0 … grad_orient_hist_7

E. Morphology — shape (8 features)

Connected-component shape stats (6): morph_n_components, morph_area_mean, morph_area_max, morph_circularity_mean, morph_eccentricity_mean, morph_solidity_mean

Top-hat / black-hat filtering (2): morph_tophat_mean, morph_blackhat_mean

F. Symmetry (5 features)

Left–right comparison (5): sym_intensity_mean_diff, sym_intensity_abs_diff, sym_intensity_std_diff, sym_hist_divergence, sym_edge_density_diff

G. Wavelet (28 features) — carried over from set D

LL approximation band (4): wav_LL_mean, wav_LL_std, wav_LL_energy, wav_LL_entropy

Per level (2 levels × 3 subbands × 4 stats = 24): wav_L{1,2}_LH_(mean/std/energy/entropy), wav_L{1,2}_HL_(…), wav_L{1,2}_HH_(…)

H. Frequency — FFT (4 features) — carried over from set D

freq_low_energy_ratio, freq_mid_energy_ratio, freq_high_energy_ratio, freq_spectral_entropy

I. Blob / lesion (5 features) — carried over from set E

blob_count, blob_mean_radius, blob_max_radius, blob_mean_contrast, blob_density

J. ROI-aware (7 features) — carried over from set E

roi_segmentation_ok, roi_intensity_mean, roi_intensity_std, roi_intensity_p90, roi_left_right_mean_diff, roi_area_frac, roi_bbox_aspect

K. GLRLM (10 features) — carried over from set F

Horizontal & vertical run-length stats (2 dirs × 5 stats): glrlm_horiz_SRE, glrlm_horiz_LRE, glrlm_horiz_GLN, glrlm_horiz_RLN, glrlm_horiz_RP, glrlm_vert_SRE, glrlm_vert_LRE, glrlm_vert_GLN, glrlm_vert_RLN, glrlm_vert_RP

SRE = short-run emphasis, LRE = long-run emphasis, GLN = gray-level non-uniformity, RLN = run-length non-uniformity, RP = run percentage.

L. Gabor (24 features) — carried over from set F

`3 frequencies (0.1, 0.25, 0.4) × 4 orientations (0°, 45°, 90°, 135°) × 2 stats (mean, std): gabor_f{freq}_t{theta}_mean, gabor_f{freq}_t{theta}_std, e.g. gabor_f0.1_t0_mean … gabor_f0.4_t2.36_std`

M. Hu moments (7 features) — carried over from set F

`Log-scaled Hu invariants of the lung mask (rotation/scale/translation invariant): hu_0 … hu_6`

N. Clinical / Anatomical (6 features) — carried over from set G

`Costophrenic angle, per side (3): cp_left_slope, cp_right_slope, cp_asymmetry`

`Lung shape/volume (3): shape_lr_area_ratio, shape_lr_height_ratio, shape_aspect`

O. Vessel / Rib Density (6 features) — new

`Overall vesselness stats (3): vessel_mean, vessel_std, vessel_density — vessel_density is the fraction of pixels inside the lung mask whose Frangi vesselness score exceeds the 90th percentile of vesselness across the whole image (a rough vessel/rib "coverage" measure).`

`Left / right split (3): vessel_left_mean, vessel_right_mean, vessel_lr_ratio — mean vesselness on each side of the midline, and their ratio (unlike Family F/N, which report a difference, this one is a ratio).`

231 raw features in total (31+48+30+12+8+5+28+4+5+7+10+24+7+6+6) are extracted per image under feature set H. This is the number produced before any cleaning — see Section 7.

4. Two versions of the image feed the features

Preprocessing produces two aligned versions of each X-ray:

- **base** — normalized only. Used for plain brightness/region stats (intensity, spatial grid, symmetry, frequency), because CLAHE would distort raw brightness comparisons.
- **texture** — CLAHE + gamma correction. Used for texture, edge, shape, wavelet, blob, GLRLM, and Gabor families, since these depend on local contrast being visible.

The vessel/rib-density family (Section 3, family O) is computed on the **base** image, not texture — the Frangi filter is designed to respond to the underlying vessel/rib structures directly, and running it on the CLAHE-enhanced image risked amplifying contrast artifacts into false vessel-like responses. This puts it alongside intensity, spatial grid, and symmetry rather than alongside the other two newest families (Hu moments, clinical/anatomical), which read the lung mask itself and are unaffected by either image version.

5. Why a lung-area mask was added

A chest X-ray contains the lungs, rib cage, spine, and black background. Bones and background do not carry disease information but still affect an average taken over the whole image. The lung-area mask (Otsu thresholding + morphology + convex hull) is now used by four families: ROI-aware (intensity/left-right stats restricted to the mask), Hu moments (shape descriptors of the mask itself), clinical/anatomical (costophrenic-angle slope and shape/volume ratios read off the mask), and — new this version — vessel/rib density, which uses the mask to restrict the vesselness percentile calculation and the midline split for the left/right features. A segmentation-quality flag (roi_segmentation_ok) is stored so unreliable segmentations can be identified rather than silently trusted by any of the four.

6. How the features connect to the spatial grid

The spatial grid family is what lets location matter to the model. Without it, two X-rays with the same overall brightness would look identical even if one has a problem confined to the top of one lung and the other is uniformly hazy. Splitting

the image into a grid and into top/middle/bottom zones lets the same intensity-style statistics be computed per region instead of only once for the whole image. This part of the pipeline is unchanged since the first version.

7. Cleaning and selecting features before modelling

Raw feature extraction now produces 231 columns for feature set H (up from 225 under set G, 219 under set F, 134 under set C). The same two cleanup steps still apply:

- **Feature cleaning** (unconditional, no label used): near-constant columns dropped (variance threshold); when two columns correlate >0.95, one of the pair is dropped as redundant. The exact final column count is dataset-dependent and is printed by clean_features().
- **Feature selection** (inside the training fold only, mutual information): for each condition separately, only the top-k features most informative for that condition are kept before scaling and training — never touching validation/test data.

8. How feature usefulness was checked

The same three diagnostics from the previous version are still run:

- **Mutual information ranking** — per-condition, per-feature top-10 bar charts, and a family-averaged radar chart.
- **PCA 2D projection** — cleaned features reduced to 2D per condition, to see whether positive cases cluster separately from normal scans.
- **Left–right symmetry check** — average left/right brightness difference, condition vs. normal, with sample size and standard error shown.

The per-condition top-10 bar charts are computed over every column in X_train, so the vessel_ columns do appear there. The family-averaged radar chart's FAMILY_PREFIXES dictionary was also updated this version to add a "Vessel/Rib Density" entry — so, unlike the clinical/anatomical family added in the previous version (which is still not represented in the radar chart), the new vessel family is included in the family-level comparison from the start. GLRLM, Gabor, Hu moments, and clinical/anatomical remain excluded from the radar chart — see the limitation below.

These checks are diagnostic, not proof of causation — high mutual information means a feature is statistically useful for telling a condition apart from normal in this dataset, not that it captures the true clinical cause.

9. Limitations to keep in mind

- Texture is still single-scale under feature set H. extract_all_features() only switches on multi-scale texture (scales=(1,2,4)) when cfg["feature_set"] is exactly "D" or "E" — feature set H falls through to the single-scale branch even though it is a superset of both. Texture family C therefore stays at 30 features, not the larger multi-scale count set D/E would otherwise produce. (Carried over from sets F and G, unchanged.)
- The family-level mutual-information radar chart still doesn't see four of the newer families. FAMILY_PREFIXES now has an entry for vessel_ (fixed this version), but glrlm_, gabor_, hu_, cp_, and shape_ still have no entry, so GLRLM, Gabor, Hu moments, and the clinical/anatomical family remain silently excluded from the family-averaged radar chart, even though their individual columns still appear in the per-condition top-10 mutual-information bar charts.

- The vessel/rib-density family's left/right fallback is a placeholder, not a missing-value flag. If a region has zero pixels above the midline split (left_mask.sum() == 0, or the same for the right side), vessel_left_mean or vessel_right_mean falls back to a fixed value (0.0) rather than NaN, which can read as "no vessel signal" instead of "segmentation gave nothing to measure" on the images where roi_segmentation_ok is already False. This is the same pattern flagged for the clinical/anatomical family in the previous version, now also present here.
- 231 is the raw, pre-cleaning count for feature set H only. It is not the number of features the models are ultimately trained on, since clean_features() and the per-label mutual-information selector both reduce it further downstream.
- Feature-to-condition mapping in the Section 2 table is a reasonable expectation based on how these conditions typically present, not a guarantee — mutual information and PCA are what actually test it on this dataset. The vessel/rib-density row is the newest and least validated of these mappings.
- Classical hand-crafted features are inherently more limited than what a CNN could learn directly from pixels; this is a constraint of the challenge rules, not a design choice.